MINI PROJECT – 1

TITLE

Student Result Processing System

Problem Statement

Develop a Python application to manage student marks, calculate percentage, assign grades, and generate academic performance reports.

Algorithm

1. Start
2. Enter student name and roll number.
3. Enter marks of five subjects.
4. Calculate total marks.
5. Calculate percentage using:

Percentage = Total Marks / 5

6. Find highest and lowest marks.
7. Assign grade based on percentage:
    - A+ : Percentage ≥ 90
    - A : Percentage ≥ 80
    - B : Percentage ≥ 70
    - C : Percentage ≥ 60
    - D : Percentage ≥ 50
    - F : Percentage < 50
8. Display student report.
9. Stop.

Python Source Code

```
print("STUDENT RESULT PROCESSING SYSTEM")


name = input("Enter Student Name: ")

roll_no = input("Enter Roll Number: ")


m1 = float(input("Enter Marks in Subject 1: "))

m2 = float(input("Enter Marks in Subject 2: "))
```

```
m3 = float(input("Enter Marks in Subject 3: "))
m4 = float(input("Enter Marks in Subject 4: "))
m5 = float(input("Enter Marks in Subject 5: "))

total = m1 + m2 + m3 + m4 + m5
percentage = total / 5

highest = max(m1, m2, m3, m4, m5)
lowest = min(m1, m2, m3, m4, m5)

if percentage >= 90:
    grade = "A+"
elif percentage >= 80:
    grade = "A"
elif percentage >= 70:
    grade = "B"
elif percentage >= 60:
    grade = "C"
elif percentage >= 50:
    grade = "D"
else:
    grade = "F"

print("\n----- STUDENT REPORT -----")
print("Student Name :", name)
print("Roll Number  :", roll_no)
print("Total Marks  :", total)
print("Percentage   :", percentage)
print("Highest Marks:", highest)
print("Lowest Marks :", lowest)
print("Grade        :", grade)
```

Sample Input

Enter Student Name: Keerthi

Enter Roll Number: 101

Enter Marks in Subject 1: 90

Enter Marks in Subject 2: 85

Enter Marks in Subject 3: 88

Enter Marks in Subject 4: 80

Enter Marks in Subject 5: 92

Sample Output

----- STUDENT REPORT -----

Student Name : Keerthi

Roll Number : 101

Total Marks : 435.0

Percentage : 87.0

Highest Marks: 92.0

Lowest Marks : 80.0

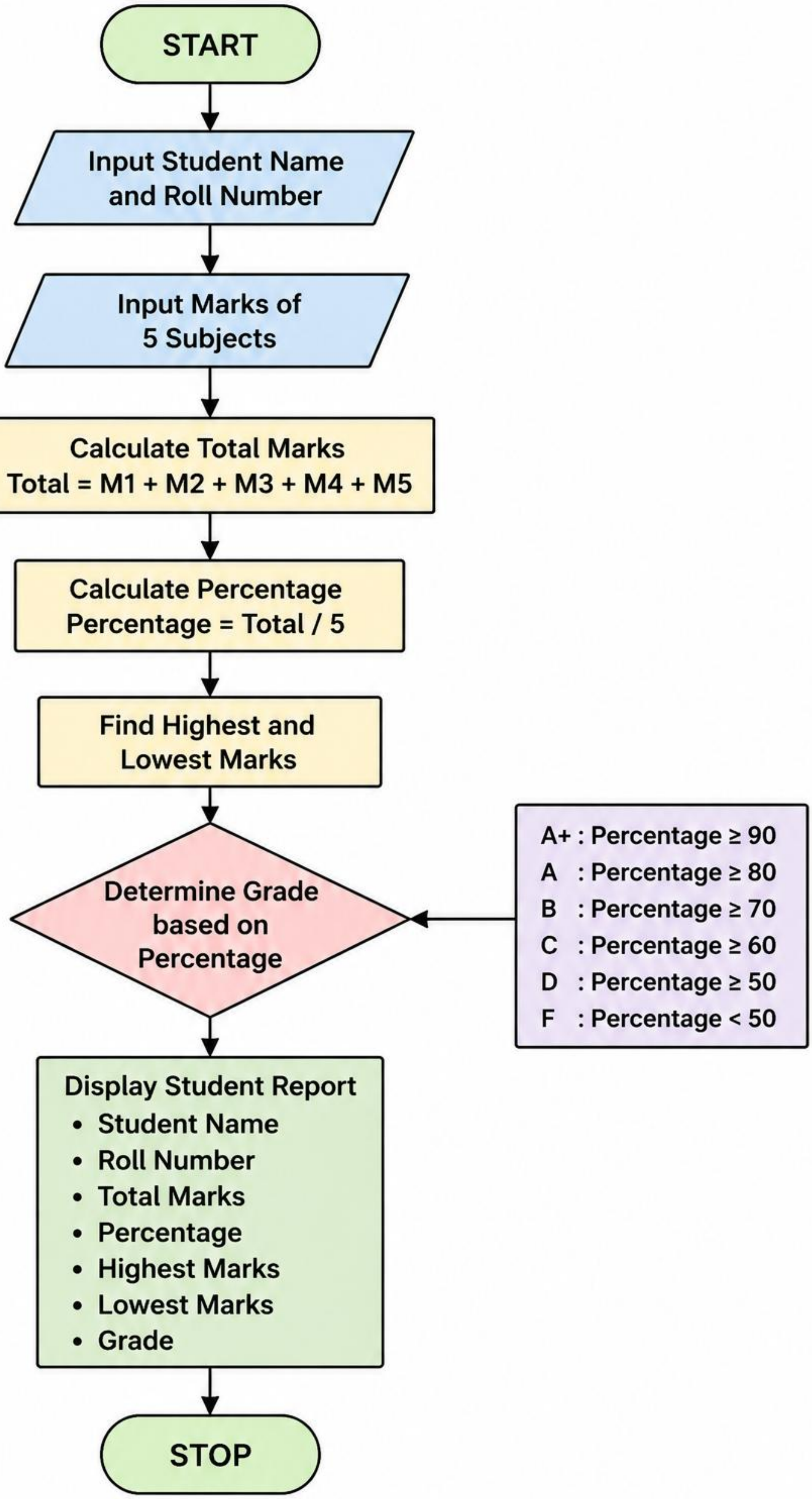
Grade : A

OUTPUT SCREENSHOT:

```
    grade = "A"
elif percentage >= 70:
    grade = "B"
elif percentage >= 60:
    grade = "C"
elif percentage >= 50:
    grade = "D"
else:
    grade = "F"

print("\n----- STUDENT REPORT -----")
print("Student Name :", name)
print("Roll Number  :", roll_no)
print("Total Marks  :", total)
print("Percentage   :", percentage)
print("Highest Marks:", highest)
print("Lowest Marks :", lowest)
print("Grade        :", grade)
```

```
STUDENT RESULT PROCESSING SYSTEM
Enter Student Name:  keerthi
Enter Roll Number:  101
Enter Marks in Subject 1:  90
Enter Marks in Subject 2:  85
Enter Marks in Subject 3:  88
Enter Marks in Subject 4:  80
Enter Marks in Subject 5:  92

----- STUDENT REPORT -----
Student Name : keerthi
Roll Number  : 101
Total Marks  : 435.0
Percentage   : 87.0
Highest Marks: 92.0
Lowest Marks : 80.0
Grade        : A
```

FLOW CHART:


START
Input Student Name and Roll Number
Input Marks of 5 Subjects
Calculate Total Marks
Total = M1 + M2 + M3 + M4 + M5
Calculate Percentage
Percentage = Total / 5
Find Highest and Lowest Marks
Determine Grade based on Percentage
A+ : Percentage ≥ 90
A : Percentage ≥ 80
B : Percentage ≥ 70
C : Percentage ≥ 60
D : Percentage ≥ 50
F : Percentage < 50
Display Student Report
• Student Name
• Roll Number
• Total Marks
• Percentage
• Highest Marks
• Lowest Marks
• Grade
STOP

README

Project Name

Student Result Processing System

Description

The Student Result Processing System is a Python-based application that accepts student details and subject marks, calculates total marks and percentage, determines grades, and generates a performance report.

Features

- Student information management
- Marks calculation
- Percentage calculation
- Grade assignment
- Highest and lowest marks identification
- Academic report generation

Tools Used

- Jupyter Notebook / VS Code

Outcome

The project successfully processes student marks and generates an academic performance report.

MINI PROJECT – 2

TITLE

Personal Finance Manager

Problem Statement

Develop a Python application to track income, expenses, savings, and monthly financial performance.

Algorithm

1. Start
2. Enter monthly income.
3. Enter house rent expense.
4. Enter food expense.
5. Enter transport expense.
6. Enter other expenses.
7. Calculate total expenses.
8. Calculate savings.
9. Calculate savings percentage.
10. Display financial report.
11. Stop.

Python Source Code

```
print("PERSONAL FINANCE MANAGER")


income = float(input("Enter Monthly Income: "))


rent = float(input("Enter House Rent Expense: "))

food = float(input("Enter Food Expense: "))

transport = float(input("Enter Transport Expense: "))

other = float(input("Enter Other Expenses: "))


total_expenses = rent + food + transport + other


savings = income - total_expenses
```

```
savings_percentage = (savings / income) * 100

print("\n----- FINANCIAL REPORT -----")
print("Monthly Income      :", income)
print("Total Expenses      :", total_expenses)
print("Total Savings       :", savings)
print("Savings Percentage  :", round(savings_percentage, 2), "%")
```

Sample Input

Enter Monthly Income: 50000

Enter House Rent Expense: 10000

Enter Food Expense: 8000

Enter Transport Expense: 3000

Enter Other Expenses: 4000

Sample Output

----- FINANCIAL REPORT -----

Monthly Income : 50000.0

Total Expenses : 25000.0

Total Savings : 25000.0

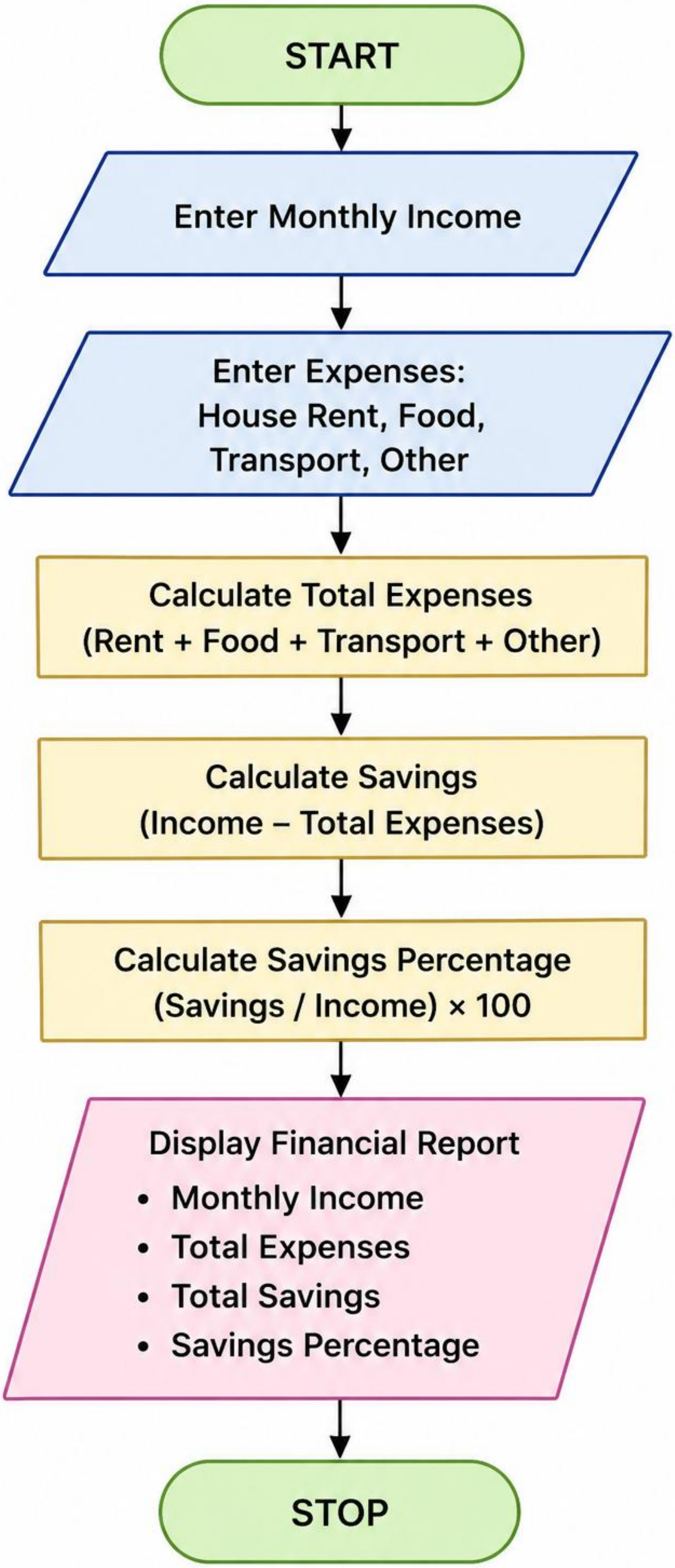
Savings Percentage : 50.0 %

OUTPUT SCREENSHOT:

```
]: print("PERSONAL FINANCE MANAGER")

income = float(input("Enter Monthly Income: "))

rent = float(input("Enter House Rent Expense: "))
food = float(input("Enter Food Expense: "))
transport = float(input("Enter Transport Expense: "))
other = float(input("Enter Other Expenses: "))

total_expenses = rent + food + transport + other

savings = income - total_expenses

savings_percentage = (savings / income) * 100

print("\n----- FINANCIAL REPORT -----")
print("Monthly Income     :", income)
print("Total Expenses     :", total_expenses)
print("Total Savings      :", savings)
print("Savings Percentage :", round(savings_percentage, 2), "%")
```

```
PERSONAL FINANCE MANAGER
Enter Monthly Income:  50000
Enter House Rent Expense:  10000
Enter Food Expense:  8000
Enter Transport Expense:  3000
Enter Other Expenses:  4000

----- FINANCIAL REPORT -----
Monthly Income     : 50000.0
Total Expenses     : 25000.0
Total Savings      : 25000.0
Savings Percentage : 50.0 %
```

FLOW CHART:


START
Enter Monthly Income
Enter Expenses:
House Rent, Food,
Transport, Other
Calculate Total Expenses
(Rent + Food + Transport + Other)
Calculate Savings
(Income – Total Expenses)
Calculate Savings Percentage
(Savings / Income) × 100
Display Financial Report
• Monthly Income
• Total Expenses
• Total Savings
• Savings Percentage
STOP

README

Project Name

Personal Finance Manager

Description

The Personal Finance Manager is a Python application used to track monthly income, expenses, and savings. It helps users understand their financial status and savings performance.

Features

- Income tracking
- Expense tracking
- Savings calculation
- Savings percentage calculation
- Monthly financial report generation

Tools Used

- Jupyter Notebook / VS Code

Outcome

The project successfully calculates monthly expenses, savings, and financial performance.